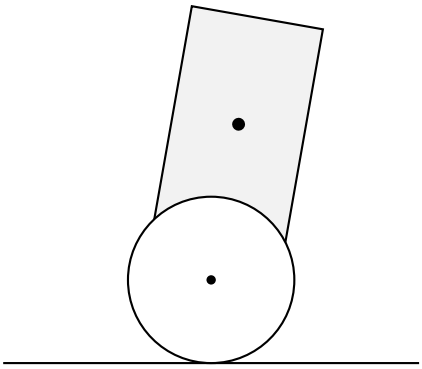
Two-Wheeled Self-Balancing Robot

Project Report



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November 1, 2026

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1 Planar dynamic model

The equations of motion of the robot in the vertical plane are derived with the Euler–Lagrange method: the kinetic and potential energy are written as functions of the generalised coordinates, the motor torque is introduced as a generalised force, and the Euler–Lagrange equations give one equation of motion per coordinate.

1.1 Assumptions

- The robot moves in a straight line on flat, horizontal ground; the two wheels move together and are modelled as a single wheel.
- The robot consists of two rigid bodies: the body and the wheel.
- The wheel rolls without slipping.
- Friction, motor dynamics and tyre compliance are neglected.
- The motor is fixed to the body. It applies a torque τ to the wheel and, by action and reaction, a torque $-\tau$ to the body.

Table 1: Parameters of the model.

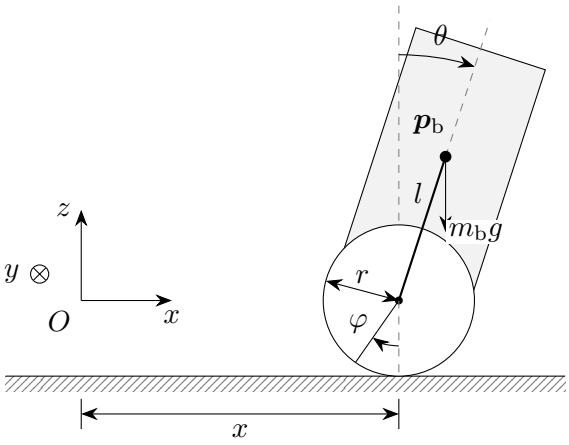
Symbol	Description	Unit
m_b	mass of the body	kg
I_b	moment of inertia of the body about its centre of mass	$\mathrm{kg\,m^2}$
l	distance between the wheel axle and the body's centre of mass	m
m_w	mass of the wheel	kg
I_w	moment of inertia of the wheel about the axle	$\mathrm{kg\,m^2}$
r	wheel radius	m
g	gravitational acceleration	$\mathrm{m/s^2}$
τ	motor torque applied to the wheel	N m

1.2 Coordinates and conventions

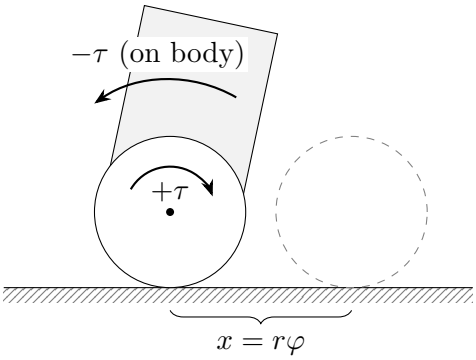
The inertial frame (Fig. 1) has the x axis horizontal, the z axis vertical and pointing up, and the origin O at the height of the wheel axle; y completes the right-handed frame and points into the page. The configuration is described by:

- x, the horizontal position of the wheel axle;
- θ, the angle of the body from the vertical, positive when the top of the body tilts towards $+x$;
- φ, the rotation angle of the wheel with respect to the ground, positive when the wheel rolls towards $+x$.

Both angles are measured in radians and are positive clockwise in Fig. 1. In the drawing $\theta > 0$ and $\varphi > 0$.



(a) Coordinates and parameters.



(b) Motor torque pair and rolling without slipping.

Figure 1: Planar model of the robot.

1.3 Rolling constraint

If the wheel does not slip, the distance travelled by the axle equals the arc length unrolled on the ground (Fig. 1b). Choosing $\varphi = 0$ when $x = 0$,

$$x = r\varphi. \tag{1}$$

Differentiating with respect to time with the product rule, and using that the radius is constant ($\dot{r} = 0$):

$$\dot{x} = \frac{\mathrm{d}(r\varphi)}{\mathrm{d}t} = \dot{r}\,\varphi + r\dot{\varphi} = r\dot{\varphi} \qquad \Longrightarrow \qquad \dot{\varphi} = \frac{\dot{x}}{r}. \tag{2}$$

The constraint expresses φ as a function of x, so the system has two degrees of freedom. The generalised coordinates are

$$\boldsymbol{q} = \begin{bmatrix} x \\ \theta \end{bmatrix}, \qquad \dot{\boldsymbol{q}} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}. \tag{3}$$

1.4 Position and velocity of the body's centre of mass

The kinetic energy requires velocities measured in the inertial frame, so the position of the body's centre of mass is written in that frame. It is the position of the axle, $(x, 0)$, plus a segment of length l tilted by θ from the vertical, whose components are $l\sin\theta$ along x and $l\cos\theta$ along z:

$$\boldsymbol{p}_\mathrm{b} = \begin{bmatrix} x + l\sin\theta \\ l\cos\theta \end{bmatrix}. \tag{4}$$

The velocity is the time derivative of each component. Both x and θ are functions of time, while l is constant:

$$\boldsymbol{v}_\mathrm{b} = \dot{\boldsymbol{p}}_\mathrm{b} = \begin{bmatrix} \dfrac{\mathrm{d}x}{\mathrm{d}t} + \dfrac{\mathrm{d}(l\sin\theta)}{\mathrm{d}t} \\ \dfrac{\mathrm{d}(l\cos\theta)}{\mathrm{d}t} \end{bmatrix}. \tag{5}$$

With the product rule, the term $\dot{l}\sin\theta$ vanishes because $\dot{l} = 0$; the chain rule then gives the derivative of a function of θ as its derivative with respect to θ times $\dot{\theta}$:

$$\frac{\mathrm{d}(l\sin\theta)}{\mathrm{d}t} = \dot{l}\sin\theta + l\,\frac{\mathrm{d}\sin\theta}{\mathrm{d}\theta}\,\frac{\mathrm{d}\theta}{\mathrm{d}t} = l\cos\theta\,\dot{\theta}, \tag{6}$$

$$\frac{\mathrm{d}(l\cos\theta)}{\mathrm{d}t} = \dot{l}\cos\theta + l\,\frac{\mathrm{d}\cos\theta}{\mathrm{d}\theta}\,\frac{\mathrm{d}\theta}{\mathrm{d}t} = -l\sin\theta\,\dot{\theta}. \tag{7}$$

Hence

$$\boldsymbol{v}_\mathrm{b} = \begin{bmatrix} \dot{x} + l\cos\theta\,\dot{\theta} \\ -l\sin\theta\,\dot{\theta} \end{bmatrix}. \tag{8}$$

1.5 Kinetic energy

The kinetic energy of a rigid body is the sum of the translational energy of its centre of mass and the rotational energy about its centre of mass, $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. For the two bodies this gives four terms.

Wheel, translation. The centre of the wheel is the axle, at $(x, 0)$, so its velocity is $\dot{x}$:

$$T_\mathrm{w,t} = \frac{1}{2}m_\mathrm{w}\dot{x}^2. \tag{9}$$

Wheel, rotation. The angular velocity of the wheel is $\dot{\varphi}$, which by Eq. (2) equals $\dot{x}/r$:

$$T_\mathrm{w,r} = \frac{1}{2}I_\mathrm{w}\dot{\varphi}^2 = \frac{1}{2}I_\mathrm{w}\left(\frac{\dot{x}}{r}\right)^2. \tag{10}$$

Body, translation. The squared speed is the sum of the squares of the two components of Eq. (8):

$$\begin{aligned} v_\mathrm{b}^2 &= (\dot{x} + l\cos\theta\,\dot{\theta})^2 + (-l\sin\theta\,\dot{\theta})^2 \\ &= \dot{x}^2 + 2l\cos\theta\,\dot{x}\dot{\theta} + l^2\cos^2\theta\,\dot{\theta}^2 + l^2\sin^2\theta\,\dot{\theta}^2 \\ &= \dot{x}^2 + 2l\cos\theta\,\dot{x}\dot{\theta} + l^2\dot{\theta}^2\left(\cos^2\theta + \sin^2\theta\right) \\ &= \dot{x}^2 + 2l\cos\theta\,\dot{x}\dot{\theta} + l^2\dot{\theta}^2, \end{aligned} \tag{11}$$

where the last step uses $\cos^2\theta + \sin^2\theta = 1$. Therefore

$$T_\mathrm{b,t} = \frac{1}{2}m_\mathrm{b}v_\mathrm{b}^2 = \frac{1}{2}m_\mathrm{b}\left(\dot{x}^2 + 2l\cos\theta\,\dot{x}\dot{\theta} + l^2\dot{\theta}^2\right). \tag{12}$$

Body, rotation. The orientation of the body is the angle θ, measured from the vertical, which is a fixed direction; the angular velocity of the body is therefore $\dot{\theta}$:

$$T_\mathrm{b,r} = \frac{1}{2}I_\mathrm{b}\dot{\theta}^2. \tag{13}$$

Total kinetic energy. Summing the four terms,

$$T = \frac{1}{2}m_{\mathrm{w}}\dot{x}^2 + \frac{1}{2}I_{\mathrm{w}}\left(\frac{\dot{x}}{r}\right)^2 + \frac{1}{2}m_{\mathrm{b}}\left(\dot{x}^2 + 2l\cos\theta\,\dot{x}\dot{\theta} + l^2\dot{\theta}^2\right) + \frac{1}{2}I_{\mathrm{b}}\dot{\theta}^2. \tag{14}$$

Expanding the body term,

$$T = \frac{1}{2}m_{\mathrm{w}}\dot{x}^2 + \frac{1}{2}\frac{I_{\mathrm{w}}}{r^2}\dot{x}^2 + \frac{1}{2}m_{\mathrm{b}}\dot{x}^2 + m_{\mathrm{b}}l\cos\theta\,\dot{x}\dot{\theta} + \frac{1}{2}m_{\mathrm{b}}l^2\dot{\theta}^2 + \frac{1}{2}I_{\mathrm{b}}\dot{\theta}^2, \tag{15}$$

and collecting the terms in $\dot{x}^2$, $\dot{x}\dot{\theta}$ and $\dot{\theta}^2$:

$$T = \frac{1}{2}\dot{x}^2\left(m_{\mathrm{w}} + \frac{I_{\mathrm{w}}}{r^2} + m_{\mathrm{b}}\right) + m_{\mathrm{b}}l\cos\theta\,\dot{x}\dot{\theta} + \frac{1}{2}\dot{\theta}^2\left(m_{\mathrm{b}}l^2 + I_{\mathrm{b}}\right). \tag{16}$$

For brevity, the three coefficients are denoted

$$a = m_{\mathrm{w}} + \frac{I_{\mathrm{w}}}{r^2} + m_{\mathrm{b}}, \qquad b = m_{\mathrm{b}}l\cos\theta, \qquad c = m_{\mathrm{b}}l^2 + I_{\mathrm{b}}, \tag{17}$$

so that $T = \frac{1}{2}a\dot{x}^2 + b\,\dot{x}\dot{\theta} + \frac{1}{2}c\,\dot{\theta}^2$.

1.6 Potential energy

The reference height $z = 0$ is the wheel axle. The centre of the wheel stays at this height on flat ground and does not contribute. The body's centre of mass is at height $l\cos\theta$, by Eq. (4):

$$V = m_{\mathrm{b}}g\,l\cos\theta. \tag{18}$$

1.7 Generalised forces

The motor torque enters the equations of motion as a generalised force Q_i for each coordinate, defined by the virtual work δW done by the motor for virtual displacements δx and $\delta\theta$:

$$\delta W = Q_x\,\delta x + Q_\theta\,\delta\theta. \tag{19}$$

The work of a torque is the torque multiplied by the rotation of the body it acts on. The motor applies $+\tau$ to the wheel, which rotates by $\delta\varphi$, and $-\tau$ to the body, which rotates by $\delta\theta$ (Fig. 1b):

$$\delta W = \tau\,\delta\varphi - \tau\,\delta\theta = \tau\,\delta(\varphi - \theta), \tag{20}$$

i.e. the work depends on the rotation of the wheel relative to the body. The virtual rotation of the wheel follows from the rolling constraint Eq. (1), with $\delta r = 0$:

$$\delta x = \delta(\varphi r) = r\,\delta\varphi + \varphi\,\delta r = r\,\delta\varphi \qquad \Longrightarrow \qquad \delta\varphi = \frac{\delta x}{r}. \tag{21}$$

Substituting,

$$\delta W = \frac{\tau}{r}\,\delta x - \tau\,\delta\theta, \tag{22}$$

and comparing with Eq. (19):

$$Q_x = \frac{\tau}{r}, \qquad Q_\theta = -\tau. \tag{23}$$

Q_x is a force (N) and Q_θ a torque (N m), consistently with x being a length and θ an angle. The contact forces between wheel and ground do no work, because the contact point does not slide; gravity is already included in V.

1.8 Euler–Lagrange equations

The Lagrangian is $L = T - V$. From Eqs. (16) and (18),

$$L = \frac{1}{2}a\dot{x}^2 + m_\mathrm{b} l \cos\theta\, \dot{x}\dot{\theta} + \frac{1}{2}c\,\dot{\theta}^2 - m_\mathrm{b} g l \cos\theta, \tag{24}$$

and for each coordinate $q_i \in \{x, \theta\}$ the Euler–Lagrange equation is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i. \tag{25}$$

Partial derivatives. Each partial derivative is taken with respect to one variable, keeping all the others constant. With respect to the velocities:

$$\frac{\partial L}{\partial \dot{x}} = a\dot{x} + m_\mathrm{b} l \cos\theta\, \dot{\theta}, \tag{26}$$

$$\frac{\partial L}{\partial \dot{\theta}} = m_\mathrm{b} l \cos\theta\, \dot{x} + c\,\dot{\theta}. \tag{27}$$

With respect to the positions: L does not contain x, and θ appears in the coupling term and in the potential energy, through $\cos\theta$:

$$\frac{\partial L}{\partial x} = 0, \tag{28}$$

$$\frac{\partial L}{\partial \theta} = -m_\mathrm{b} l \sin\theta\, \dot{x}\dot{\theta} + m_\mathrm{b} g l \sin\theta. \tag{29}$$

Time derivatives. The total time derivative follows the motion, so every quantity that depends on time (x, θ, $\dot{x}$, $\dot{\theta}$) is differentiated; only the parameters are constant. In Eq. (26), a is constant, so $\frac{\mathrm{d}}{\mathrm{d}t}(a\dot{x}) = a\ddot{x}$; the second term is the product of the two functions of time $\cos\theta$ and $\dot{\theta}$, and $\frac{\mathrm{d}}{\mathrm{d}t}\cos\theta = -\sin\theta\,\dot{\theta}$:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} &= a\ddot{x} + m_\mathrm{b} l \left(-\sin\theta\,\dot{\theta}\cdot\dot{\theta} + \cos\theta\,\ddot{\theta}\right) \\ &= a\ddot{x} - m_\mathrm{b} l \sin\theta\,\dot{\theta}^2 + m_\mathrm{b} l \cos\theta\,\ddot{\theta}.\end{aligned} \tag{30}$$

In the same way, from Eq. (27):

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\theta}} &= m_\mathrm{b} l \left(-\sin\theta\,\dot{\theta}\,\dot{x} + \cos\theta\,\ddot{x}\right) + c\,\ddot{\theta} \\ &= -m_\mathrm{b} l \sin\theta\,\dot{\theta}\dot{x} + m_\mathrm{b} l \cos\theta\,\ddot{x} + c\,\ddot{\theta}.\end{aligned} \tag{31}$$

Equation of motion for x. Substituting into Eq. (25) with $Q_x = \tau/r$:

$$a\ddot{x} - m_\mathrm{b} l \sin\theta\,\dot{\theta}^2 + m_\mathrm{b} l \cos\theta\,\ddot{\theta} - 0 = \frac{\tau}{r}. \tag{32}$$

Equation of motion for θ. Substituting into Eq. (25) with $Q_\theta = -\tau$, the minus sign in front of $\partial L/\partial\theta$ applies to both terms of Eq. (29):

$$\begin{aligned}-m_\mathrm{b} l \sin\theta\,\dot{\theta}\dot{x} + m_\mathrm{b} l \cos\theta\,\ddot{x} + c\,\ddot{\theta} - \left(-m_\mathrm{b} l \sin\theta\,\dot{x}\dot{\theta} + m_\mathrm{b} g l \sin\theta\right) &= -\tau \\ -m_\mathrm{b} l \sin\theta\,\dot{\theta}\dot{x} + m_\mathrm{b} l \cos\theta\,\ddot{x} + c\,\ddot{\theta} + m_\mathrm{b} l \sin\theta\,\dot{x}\dot{\theta} - m_\mathrm{b} g l \sin\theta &= -\tau.\end{aligned} \tag{33}$$

The first and fourth terms cancel, leaving

$$m_\mathrm{b} l \cos\theta\,\ddot{x} + c\,\ddot{\theta} - m_\mathrm{b} g l \sin\theta = -\tau. \tag{34}$$

Summary. With a and c from Eq. (17), the equations of motion are

$$a\ddot{x} + m_\mathrm{b} l \cos\theta\, \ddot{\theta} - m_\mathrm{b} l \sin\theta\, \dot{\theta}^2 = \frac{\tau}{r}, \tag{35}$$

$$m_\mathrm{b} l \cos\theta\, \ddot{x} + c\, \ddot{\theta} - m_\mathrm{b} g l \sin\theta = -\tau. \tag{36}$$

1.9 Matrix form

Eqs. (35) and (36) are written in the form

$$\boldsymbol{M}(\boldsymbol{q})\, \ddot{\boldsymbol{q}} + \boldsymbol{h}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{B}\, \tau, \tag{37}$$

where row 1 corresponds to the equation of x and row 2 to the equation of θ. In each equation, the coefficients of the accelerations $\ddot{x}$ and $\ddot{\theta}$ form the corresponding row of the mass matrix $\boldsymbol{M}$; the remaining terms on the left-hand side form the vector $\boldsymbol{h}$; the coefficients of τ on the right-hand side form the vector $\boldsymbol{B}$:

$$\underbrace{\begin{bmatrix} m_\mathrm{w} + \dfrac{I_\mathrm{w}}{r^2} + m_\mathrm{b} & m_\mathrm{b} l \cos\theta \\ m_\mathrm{b} l \cos\theta & m_\mathrm{b} l^2 + I_\mathrm{b} \end{bmatrix}}_{\boldsymbol{M}(\boldsymbol{q})} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \underbrace{\begin{bmatrix} -m_\mathrm{b} l \sin\theta\, \dot{\theta}^2 \\ -m_\mathrm{b} g l \sin\theta \end{bmatrix}}_{\boldsymbol{h}(\boldsymbol{q},\dot{\boldsymbol{q}})} = \underbrace{\begin{bmatrix} 1/r \\ -1 \end{bmatrix}}_{\boldsymbol{B}} \tau. \tag{38}$$

The same mass matrix follows directly from the kinetic energy. Writing $T = \frac{1}{2}\dot{\boldsymbol{q}}^\mathsf{T} \boldsymbol{M} \dot{\boldsymbol{q}}$ for a generic 2×2 matrix $\boldsymbol{M}$,

$$\frac{1}{2}\dot{\boldsymbol{q}}^\mathsf{T} \boldsymbol{M} \dot{\boldsymbol{q}} = \frac{1}{2}\left(M_{11}\dot{x}^2 + (M_{12} + M_{21})\, \dot{x}\dot{\theta} + M_{22}\dot{\theta}^2 \right), \tag{39}$$

and comparing with $T = \frac{1}{2}a\dot{x}^2 + b\, \dot{x}\dot{\theta} + \frac{1}{2}c\, \dot{\theta}^2$ gives $M_{11} = a$, $M_{22} = c$ and, for a symmetric matrix, $M_{12} = M_{21} = b$, in agreement with Eq. (38).

1.10 Properties of the model

Mass matrix. $\boldsymbol{M}$ is symmetric, $\boldsymbol{M}^\mathsf{T} = \boldsymbol{M}$. Its diagonal entries a and c are constant; only the off-diagonal entry $b = m_\mathrm{b} l \cos\theta$ depends on the configuration. Its determinant,

$$\det \boldsymbol{M} = ac - b^2 = \left(m_\mathrm{w} + \frac{I_\mathrm{w}}{r^2} \right)\left(m_\mathrm{b} l^2 + I_\mathrm{b} \right) + m_\mathrm{b} I_\mathrm{b} + m_\mathrm{b}^2 l^2 \sin^2\theta, \tag{40}$$

is positive for every θ, so $\boldsymbol{M}$ is always invertible.

Equilibrium. For the robot upright and at rest with no input ($\theta = 0$, $\dot{x} = \dot{\theta} = 0$, $\tau = 0$):

$$h_1 = -m_\mathrm{b} l \sin 0 \cdot 0^2 = 0, \qquad h_2 = -m_\mathrm{b} g l \sin 0 = 0, \qquad \boldsymbol{B}\tau = \boldsymbol{0}. \tag{41}$$

Eq. (37) reduces to $\boldsymbol{M}\ddot{\boldsymbol{q}} = \boldsymbol{0}$ and, since $\boldsymbol{M}$ is invertible, $\ddot{\boldsymbol{q}} = \boldsymbol{0}$: the upright configuration at rest is an equilibrium.

Stability of the equilibrium. With the robot at rest, $\tau = 0$ and a small tilt $\theta > 0$, Eq. (37) gives $\boldsymbol{M}\ddot{\boldsymbol{q}} = -\boldsymbol{h} = \begin{bmatrix} 0 & m_\mathrm{b} g l \sin\theta \end{bmatrix}^\mathsf{T}$, hence

$$\ddot{\theta} = \frac{a\, m_\mathrm{b} g l \sin\theta}{\det \boldsymbol{M}} > 0. \tag{42}$$

The angular acceleration has the same sign as the tilt, so the tilt grows: the equilibrium is unstable. Consistently, the potential energy Eq. (18) is maximum at $\theta = 0$.

Underactuation. $\boldsymbol{B} = \begin{bmatrix} 1/r & -1 \end{bmatrix}^{\mathsf{T}}$ is a single non-zero column, so $\operatorname{rank} \boldsymbol{B} = 1$, while the system has two degrees of freedom. The input τ always acts on both coordinates in the fixed ratio $Q_x : Q_\theta = 1/r : -1$, and $\ddot{x}$ and $\ddot{\theta}$ cannot be assigned independently: the system is underactuated.

2 Numerical simulation

The model of Section 1 was implemented in Python with NumPy only (package `balancebot`), integrated numerically, verified with a set of automated tests, and used for a first experiment: the free fall of the robot with the motor off.

2.1 State-space form

The equations of motion are second order in $\boldsymbol{q}$. Stacking positions and velocities into the state

$$\boldsymbol{s} = \begin{bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{bmatrix} = \begin{bmatrix} x & \theta & \dot{x} & \dot{\theta} \end{bmatrix}^{\mathsf{T}} \tag{43}$$

turns them into a first-order system $\dot{\boldsymbol{s}} = \boldsymbol{f}(\boldsymbol{s}, \tau)$. The first two components of $\dot{\boldsymbol{s}}$ are the velocities, already contained in $\boldsymbol{s}$; the last two are the accelerations, obtained from Eq. (37):

$$\boldsymbol{f}(\boldsymbol{s}, \tau) = \begin{bmatrix} \dot{\boldsymbol{q}} \\ \ddot{\boldsymbol{q}} \end{bmatrix}, \qquad \boldsymbol{M}(\boldsymbol{q})\, \ddot{\boldsymbol{q}} = \boldsymbol{B}\tau - \boldsymbol{h}(\boldsymbol{q}, \dot{\boldsymbol{q}}). \tag{44}$$

The accelerations are computed by solving this 2×2 linear system rather than by inverting $\boldsymbol{M}$; the system always has a unique solution because $\det \boldsymbol{M} > 0$ (Section 1). In the simulation loop the torque is computed once per step and held constant during the step (zero-order hold), as a digital controller does.

2.2 Numerical integration

Eq. (44) has no closed-form solution, so the state is advanced in steps of length Δt. Two integrators were implemented.

Explicit Euler. The rate of change at the beginning of the step is assumed constant over the whole step:

$$\boldsymbol{s}_{k+1} = \boldsymbol{s}_k + \Delta t\, \boldsymbol{f}(\boldsymbol{s}_k). \tag{45}$$

Fourth-order Runge–Kutta (RK4). The rate of change is evaluated four times inside the step, at the start ($\boldsymbol{k}_1$), twice at mid-step ($\boldsymbol{k}_2$, $\boldsymbol{k}_3$) and at the end ($\boldsymbol{k}_4$), and averaged with weights $1, 2, 2, 1$:

$$\begin{aligned} \boldsymbol{k}_1 &= \boldsymbol{f}(\boldsymbol{s}_k), & \boldsymbol{k}_2 &= \boldsymbol{f}\left(\boldsymbol{s}_k + \tfrac{\Delta t}{2}\boldsymbol{k}_1\right), \\ \boldsymbol{k}_3 &= \boldsymbol{f}\left(\boldsymbol{s}_k + \tfrac{\Delta t}{2}\boldsymbol{k}_2\right), & \boldsymbol{k}_4 &= \boldsymbol{f}\left(\boldsymbol{s}_k + \Delta t\, \boldsymbol{k}_3\right), \end{aligned} \qquad \boldsymbol{s}_{k+1} = \boldsymbol{s}_k + \frac{\Delta t}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right). \tag{46}$$

Order of accuracy. Both integrators were tested on the harmonic oscillator $\ddot{y} = -\omega^2 y$ with $\omega = 2$, $y(0) = 1$, $\dot{y}(0) = 0$, whose exact solution is $y(t) = \cos \omega t$. The error at $t = 2$ was computed for two step sizes (Table 2). Halving the step divides the error by 2.00 for Euler and by 15.76 for RK4, in agreement with first-order (2^1) and fourth-order ($2^4 = 16$) accuracy.

Table 2: Error at $t = 2$ on the harmonic oscillator.

Integrator	Δt	Error	Ratio
Euler	1×10^{-3}	2.624×10^{-3}	2.00
	5×10^{-4}	1.310×10^{-3}	
RK4	2×10^{-2}	6.268×10^{-8}	15.76
	1×10^{-2}	3.978×10^{-9}	

2.3 Verification of the implementation

The implementation is checked by 28 automated tests (`pytest`), each derived from a property of the model:

- $\boldsymbol{M}$ is symmetric and positive definite for $\theta \in \{0°, \pm 5°, \pm 30°, \pm 80°\}$, and does not depend on x;
- $\boldsymbol{M}$, $\boldsymbol{h}$ and $\boldsymbol{B}$ coincide numerically with those obtained from an independent symbolic derivation (SymPy) at the same values of θ;
- the first two components of $\dot{\boldsymbol{s}}$ equal the velocities, and the upright state at rest with $\tau = 0$ gives $\dot{\boldsymbol{s}} = \boldsymbol{0}$ (equilibrium);
- from rest with $\theta = 5°$ and $\tau = 0$: $\ddot{\theta} > 0$ and $\ddot{x} < 0$; from the upright state with $\tau > 0$: $\ddot{x} > 0$ and $\ddot{\theta} < 0$, as predicted by the signs of $\boldsymbol{B}$;
- the energy computed from $\boldsymbol{M}$ satisfies $T = \frac{1}{2}\dot{\boldsymbol{q}}^{\mathsf{T}}\boldsymbol{M}\dot{\boldsymbol{q}}$, and $V = m_{\mathrm{b}} g l$ at $\theta = 0$, $V = 0$ at $\theta = 90°$;
- with $\tau = 0$, RK4 and $\Delta t = 1\,\mathrm{ms}$, the total energy $T + V$ is constant over 2 s to a relative error below 1×10^{-6};
- one Euler step matches the value computed by hand, RK4 integrates $\dot{s} = t^3$ exactly, and both have the order of Table 2.

2.4 Free fall with the motor off

Set-up. The parameters are nominal estimates (Table 3), to be replaced by measured values. The robot starts at rest with $\theta_0 = 3°$ and $\tau = 0$, and is integrated with $\Delta t = 5\,\mathrm{ms}$. The simulation stops at the first step with $|\theta| \geq 90°$, when the body reaches the ground: the model has no contact between body and ground, so the motion beyond that point is not physical.

Table 3: Nominal parameters used in the simulation.

Parameter	Value	Basis of the estimate
m_{b}	$0.55\,\mathrm{kg}$	frame, motors, battery, electronics
I_{b}	$1.7 \times 10^{-3}\,\mathrm{kg\,m^2}$	box $0.18\,\mathrm{m} \times 0.06\,\mathrm{m}$, $m_{\mathrm{b}}(h^2 + d^2)/12$
l	$0.07\,\mathrm{m}$	
m_{w}	$0.06\,\mathrm{kg}$	two wheels
r	$0.0325\,\mathrm{m}$	65 mm wheels
I_{w}	$5.0 \times 10^{-5}\,\mathrm{kg\,m^2}$	$\approx 0.75\, m_{\mathrm{w}} r^2$
g	$9.81\,\mathrm{m/s^2}$	

Results. The body reaches the horizontal after 0.345 s (Figs. 2 and 3). The tilt grows slowly at first and then rapidly: 3° at $t = 0$, 11° at 0.15 s, 37° at 0.25 s, 62° at 0.30 s, as expected from the unstable equilibrium of Section 1. While the body falls forward, the wheels roll backwards, by 5.55 cm in total; their speed first grows and then returns to zero as the body approaches the horizontal.

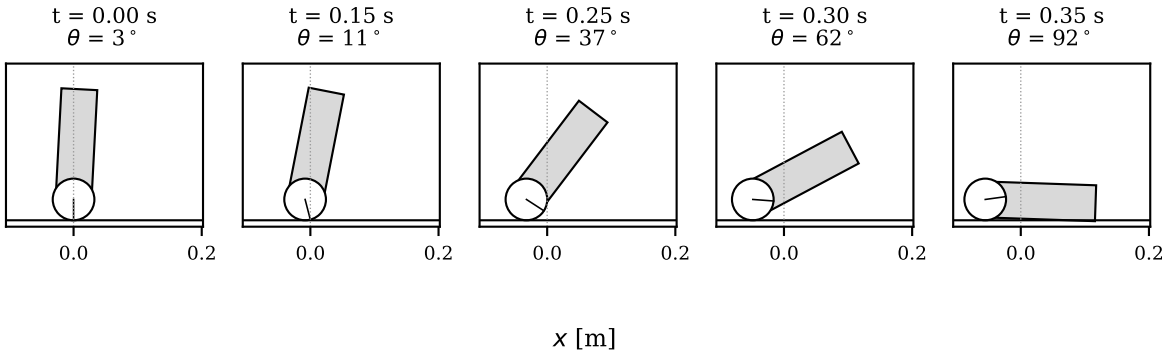


Figure 2: Free fall with the motor off (RK4, $\Delta t = 5$ ms). The dotted line marks the initial position of the axle.

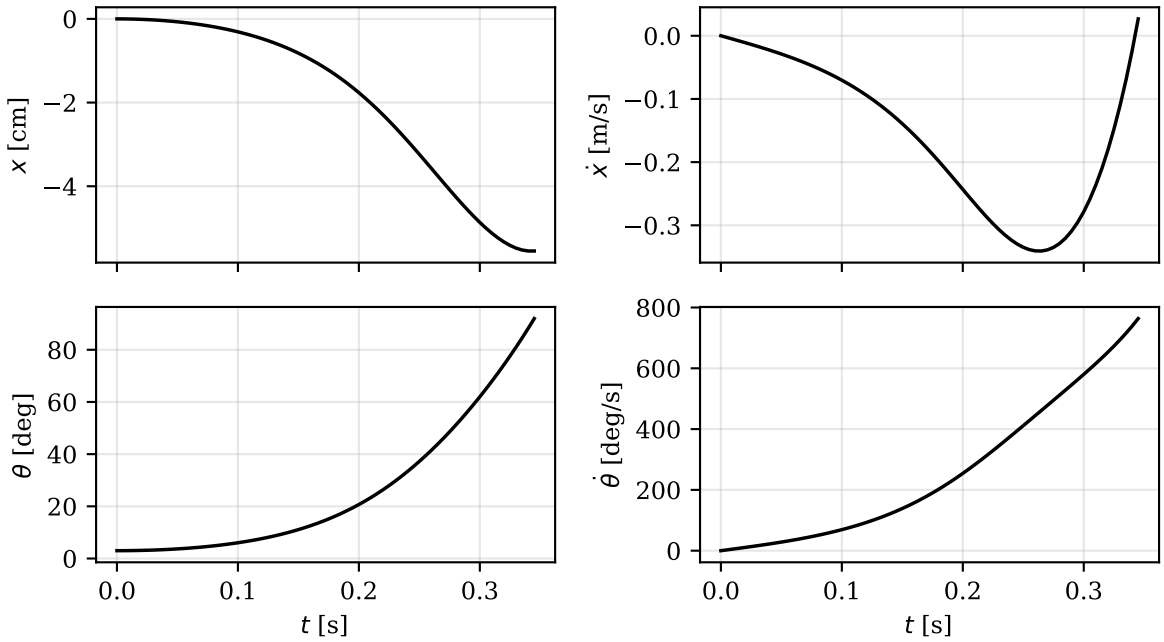


Figure 3: State during the free fall (RK4, $\Delta t = 5$ ms).

The motion of the wheels follows from the equation of motion of x. With $\tau = 0$, $Q_x = 0$ and $\partial L/\partial x = 0$, so Eq. (25) reduces to $\frac{\mathrm{d}}{\mathrm{d}t}\left(\partial L/\partial \dot{x}\right) = 0$: the generalised momentum

$$p_x = \frac{\partial L}{\partial \dot{x}} = a\,\dot{x} + m_\mathrm{b} l \cos\theta\,\dot{\theta} \tag{47}$$

is conserved. Starting from rest, $p_x = 0$ at all times, hence

$$\dot{x} = -\frac{m_\mathrm{b} l \cos\theta}{a}\,\dot{\theta}. \tag{48}$$

A forward rotation of the body ($\dot{\theta} > 0$) therefore drives the wheels backwards, through the coupling term $b = m_\mathrm{b} l \cos\theta$, and the effect vanishes at $\theta = 90°$, where $\cos\theta = 0$. Along the simulated trajectory $|p_x|$ stays below $1.7 \times 10^{-8}\,\mathrm{kg\,m/s}$.

Energy. Figure 4 shows the relative drift of the total energy for the two integrators with the same step. The maximum drift is 6.2×10^{-3} with explicit Euler and 7.6×10^{-8} with RK4. The Euler error grows as the motion accelerates; RK4 conserves the energy to within the accuracy required by the tests, and is the default integrator of the simulator.

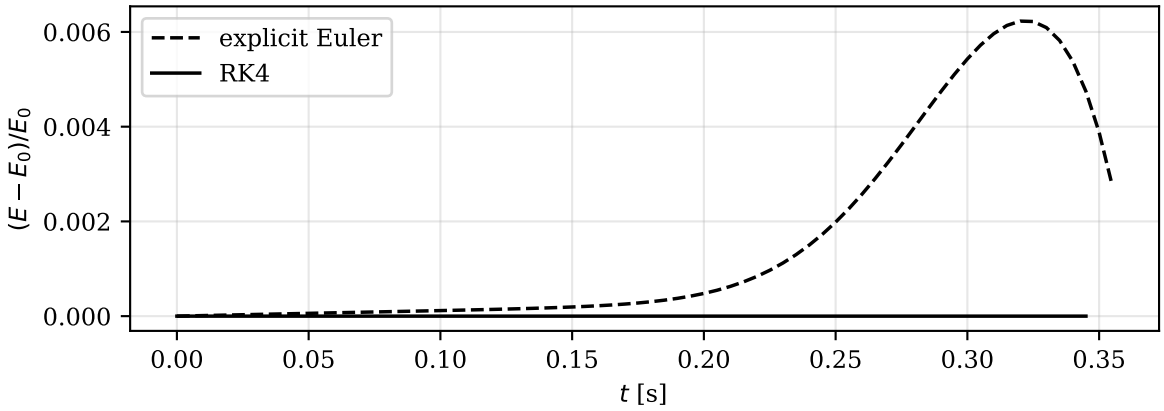


Figure 4: Relative drift of the total energy $E = T + V$ during the free fall, explicit Euler and RK4, $\Delta t = 5\,\mathrm{ms}$.

3 Linearisation about the upright equilibrium

The equations of motion Eqs. (35) and (36) are nonlinear in θ and $\dot{\theta}$. Near the upright equilibrium they can be replaced by a linear time-invariant model

$$\dot{\boldsymbol{s}} = \boldsymbol{A}\,\boldsymbol{s} + \boldsymbol{B}_\mathrm{lin}\,\tau, \qquad \boldsymbol{s} = \begin{bmatrix} x & \theta & \dot{x} & \dot{\theta} \end{bmatrix}^\mathsf{T}, \tag{49}$$

with constant matrices $\boldsymbol{A} \in \mathbb{R}^{4\times4}$ and $\boldsymbol{B}_\mathrm{lin} \in \mathbb{R}^{4\times1}$. A model of this form is introduced because its behaviour is fully characterised by the two matrices: stability follows from the eigenvalues of $\boldsymbol{A}$, controllability from a rank condition, and the standard methods of linear control design apply to it directly. None of these tools applies to the nonlinear model.

3.1 First-order approximation

The equilibrium is $\theta = 0$, $\dot{x} = \dot{\theta} = 0$, $\tau = 0$, for any x (Section 1). Near it θ, $\dot{x}$, $\dot{\theta}$ and τ are small quantities, and each term of the equations is kept only up to first order in them. The trigonometric functions are replaced by their first-order Taylor expansion about $\theta_0 = 0$,

$$f(\theta) \approx f(0) + f'(0)\,\theta, \tag{50}$$

which gives

$$\sin\theta \approx \sin 0 + \cos 0 \cdot \theta = \theta, \qquad \cos\theta \approx \cos 0 - \sin 0 \cdot \theta = 1. \tag{51}$$

Term by term:

- $m_\mathrm{b} l \cos\theta\,\ddot{\theta} \approx m_\mathrm{b} l\,\ddot{\theta}$: the correction to $\cos\theta \approx 1$ is of order θ^2 and multiplies $\ddot{\theta}$, so it is of third order;

- $-m_\mathrm{b} l \sin\theta\, \dot\theta^2 \approx -m_\mathrm{b} l\, \theta\, \dot\theta^2$: product of three small quantities, third order, neglected;
- $-m_\mathrm{b} g l \sin\theta \approx -m_\mathrm{b} g l\, \theta$: first order, kept.

With $b_0 = m_\mathrm{b} l$, the value of $b = m_\mathrm{b} l \cos\theta$ at $\theta = 0$, the linearised equations of motion are

$$\underbrace{\begin{bmatrix} a & b_0 \\ b_0 & c \end{bmatrix}}_{\boldsymbol{M}_0} \begin{bmatrix} \ddot x \\ \ddot\theta \end{bmatrix} + \begin{bmatrix} 0 \\ -m_\mathrm{b} g l\, \theta \end{bmatrix} = \begin{bmatrix} 1/r \\ -1 \end{bmatrix} \tau, \tag{52}$$

with a and c from Eq. (17).

3.2 State-space matrices

Moving the gravity term to the right-hand side of Eq. (52),

$$\boldsymbol{M}_0 \begin{bmatrix} \ddot x \\ \ddot\theta \end{bmatrix} = \begin{bmatrix} \tau/r \\ -\tau + m_\mathrm{b} g l\, \theta \end{bmatrix}. \tag{53}$$

The inverse of $\boldsymbol{M}_0$ is

$$\boldsymbol{M}_0^{-1} = \frac{1}{D} \begin{bmatrix} c & -b_0 \\ -b_0 & a \end{bmatrix}, \qquad D = \det \boldsymbol{M}_0 = ac - b_0^2 > 0, \tag{54}$$

so that

$$\ddot x = \frac{1}{D}\left[c\,\frac{\tau}{r} - b_0\,(-\tau + m_\mathrm{b} g l\, \theta) \right] = \frac{1}{D}\left[-b_0\, m_\mathrm{b} g l\, \theta + \left(\frac{c}{r} + b_0\right)\tau \right], \tag{55}$$

$$\ddot\theta = \frac{1}{D}\left[-b_0\,\frac{\tau}{r} + a\,(-\tau + m_\mathrm{b} g l\, \theta) \right] = \frac{1}{D}\left[a\, m_\mathrm{b} g l\, \theta - \left(a + \frac{b_0}{r}\right)\tau \right]. \tag{56}$$

The first two components of $\dot{\boldsymbol{s}}$ are $\dot x$ and $\dot\theta$, the last two are Eqs. (55) and (56). Writing $\alpha = -b_0 m_\mathrm{b} g l / D$ and $\beta = a m_\mathrm{b} g l / D$,

$$\boldsymbol{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \alpha & 0 & 0 \\ 0 & \beta & 0 & 0 \end{bmatrix}, \qquad \boldsymbol{B}_\mathrm{lin} = \begin{bmatrix} 0 \\ 0 \\ b_3 \\ b_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ (c/r + b_0)/D \\ -(a + b_0/r)/D \end{bmatrix}. \tag{57}$$

With the parameters of Table 3: $\alpha = -10.34\,\mathrm{s}^{-2}$, $\beta = 176.48\,\mathrm{s}^{-2}$, $b_3 = 123.50\,\mathrm{kg}^{-1}\,\mathrm{m}^{-1}$, $b_4 = -1309.37\,\mathrm{kg}^{-1}\,\mathrm{m}^{-2}$. The same matrices were obtained numerically, by central finite differences of the nonlinear $\boldsymbol{f}(\boldsymbol{s}, \tau)$ of Eq. (44) about the equilibrium with step 1×10^{-6}; the largest difference between the two is 2.2×10^{-10}.

3.3 Eigenvalues

The eigenvalues of $\boldsymbol{A}$ are the roots of $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$:

$$\det \begin{bmatrix} \lambda & 0 & -1 & 0 \\ 0 & \lambda & 0 & -1 \\ 0 & -\alpha & \lambda & 0 \\ 0 & -\beta & 0 & \lambda \end{bmatrix}. \tag{58}$$

The first column has a single non-zero entry, so expanding along it,

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \lambda \det \begin{bmatrix} \lambda & 0 & -1 \\ -\alpha & \lambda & 0 \\ -\beta & 0 & \lambda \end{bmatrix} = \lambda \left[\lambda \cdot \lambda^2 - 1 \cdot (0 + \lambda\beta) \right] = \lambda^2 \left(\lambda^2 - \beta \right). \tag{59}$$

The eigenvalues are therefore

$$\lambda_{1,2} = 0, \qquad \lambda_{3,4} = \pm\sqrt{\beta} = \pm\sqrt{\frac{a\, m_{\mathrm{b}} g l}{D}} = \pm 13.28\,\mathrm{s}^{-1}. \tag{60}$$

The double zero belongs to the position: with $\tau = 0$ nothing acts on x other than the coupling with θ, and the robot has no preferred position. The positive eigenvalue makes the equilibrium unstable: a deviation in θ grows as $e^{\sqrt{\beta}\,t}$, by a factor e every $1/\sqrt{\beta} = 75\,\mathrm{ms}$. The coefficient α does not affect the eigenvalues: the motion of the wheels does not act back on θ.

Comparison with the nonlinear model. With $\tau = 0$, Eq. (56) reduces to $\ddot{\theta} = \beta\,\theta$, whose general solution is $\theta(t) = C_1 e^{\sqrt{\beta}t} + C_2 e^{-\sqrt{\beta}t}$. From rest, $\theta(0) = \theta_0$ and $\dot{\theta}(0) = 0$ give $C_1 = C_2 = \theta_0/2$, hence

$$\theta(t) = \theta_0 \cosh\!\left(\sqrt{\beta}\,t\right). \tag{61}$$

Table 4 compares this solution with the nonlinear simulation of Section 2 (RK4, $\Delta t = 1\,\mathrm{ms}$), both from $\theta_0 = 3°$.

Table 4: Free fall from $\theta_0 = 3°$: linearised and nonlinear model.

t [s]	θ, linear [deg]	θ, nonlinear [deg]
0.05	3.7	3.7
0.10	6.1	6.0
0.15	11.2	11.1
0.20	21.5	20.7
0.25	41.6	37.2
0.30	80.7	62.0

Limits of the linear model. The two models agree within 1° up to about 20° and diverge beyond. The linear model uses θ in place of $\sin\theta$; since $\theta > \sin\theta$ for $\theta > 0$, it overestimates the gravitational torque and predicts a faster fall. The relative error of the approximation is 1% at 14° and 4.7% at 30°, and the neglected terms (such as $\theta\dot{\theta}^2$) also grow with the speed of the motion. The linear model is therefore a valid description only in a neighbourhood of the equilibrium, and results obtained from it must be checked on the nonlinear model.

3.4 Controllability

The system Eq. (49) is controllable, i.e. the single input τ can bring the state from any initial value to any final value in finite time, if and only if the controllability matrix

$$\boldsymbol{C} = \begin{bmatrix} \boldsymbol{B}_{\mathrm{lin}} & \boldsymbol{A}\boldsymbol{B}_{\mathrm{lin}} & \boldsymbol{A}^2\boldsymbol{B}_{\mathrm{lin}} & \boldsymbol{A}^3\boldsymbol{B}_{\mathrm{lin}} \end{bmatrix} \tag{62}$$

has rank 4. From Eq. (57),

$$\boldsymbol{B}_{\mathrm{lin}} = \begin{bmatrix} 0 \\ 0 \\ b_3 \\ b_4 \end{bmatrix}, \quad \boldsymbol{A}\boldsymbol{B}_{\mathrm{lin}} = \begin{bmatrix} b_3 \\ b_4 \\ 0 \\ 0 \end{bmatrix}, \quad \boldsymbol{A}^2\boldsymbol{B}_{\mathrm{lin}} = \begin{bmatrix} 0 \\ 0 \\ \alpha b_4 \\ \beta b_4 \end{bmatrix}, \quad \boldsymbol{A}^3\boldsymbol{B}_{\mathrm{lin}} = \begin{bmatrix} \alpha b_4 \\ \beta b_4 \\ 0 \\ 0 \end{bmatrix}. \tag{63}$$

Reordering the columns as (2, 4, 1, 3), which changes at most the sign of the determinant, gives a block-diagonal matrix with two equal blocks

$$\boldsymbol{G} = \begin{bmatrix} b_3 & \alpha b_4 \\ b_4 & \beta b_4 \end{bmatrix}, \qquad |\det \boldsymbol{C}| = (\det \boldsymbol{G})^2, \tag{64}$$

with

$$\begin{aligned}\det \boldsymbol{G} &= b_4 \left(\beta b_3 - \alpha b_4\right) = b_4 \frac{m_\mathrm{b} g l}{D^2} \left[a \left(\frac{c}{r} + b_0\right) - b_0 \left(a + \frac{b_0}{r}\right)\right] \\ &= b_4 \frac{m_\mathrm{b} g l}{D^2} \frac{ac - b_0^2}{r} = b_4 \frac{m_\mathrm{b} g l}{rD}.\end{aligned} \tag{65}$$

Since $b_4 \neq 0$ and $D > 0$, $\det \boldsymbol{C} \neq 0$ for every value of the parameters: the linearised system is controllable, although it has two degrees of freedom and a single input. The rank was also confirmed numerically (`numpy.linalg.matrix_rank`) with the parameters of Table 3.